

Introduction to CV

Today's Agenda

Definition and application

Light and Color

Primary Colors

Colors on Displays

Storing Images

Q&A and Coding

Computer vision

It is an area of machine learning that is related to

- Recording digital images and videos
- Saving the results
- Editing
- Interpretation



Computer vision

Application areas

- Healthcare & Medical Imaging
- Autonomous Vehicles & Transportation
- Security & Surveillance
- Retail & ECommerce
- Agriculture & Farming
- Entertainment & Media
- Banking & Finance

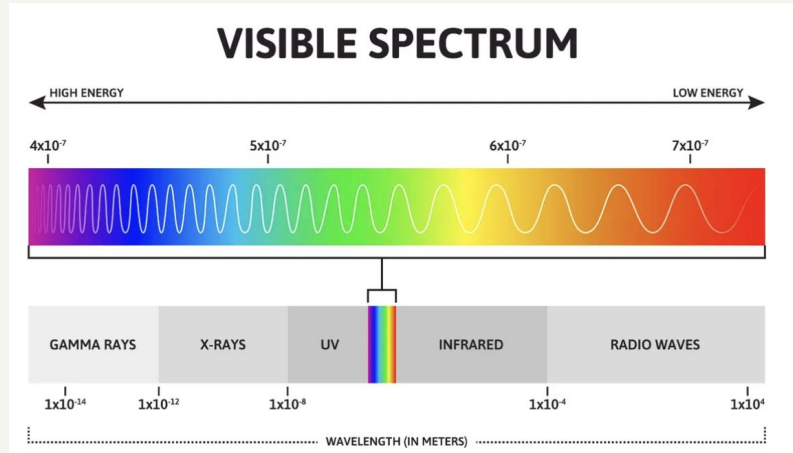
Breakout rooms

Use 5 - 10 minutes to discuss where you have used CV

Light and Color

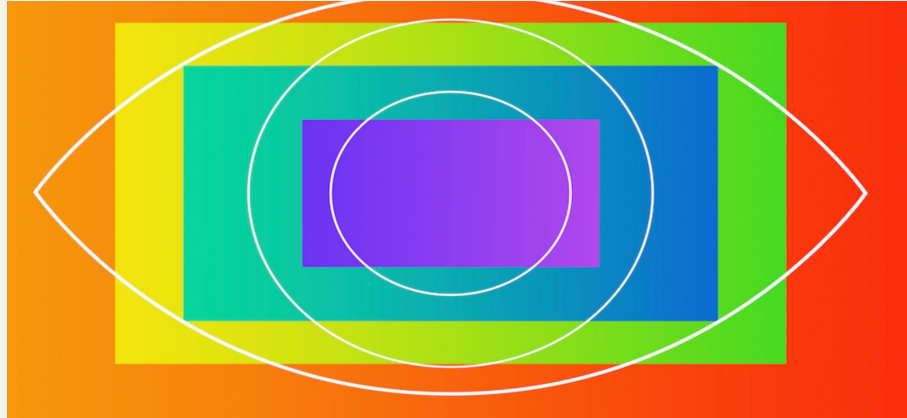
Visible Light

- It is a **small part of the electromagnetic spectrum** that can be detected by the human eye.
- The electromagnetic spectrum includes a wide range of wavelengths, but only a narrow range (approximately **380 to 750 nanometers**) is visible to us.



Color Perception

- The color of an object depends on the **wavelengths of light it reflects** and how our eyes interpret those wavelengths.
- Different objects **absorb some wavelengths** and **reflect others**.
- The wavelengths that are reflected determine the **color we perceive**.



Color Perception

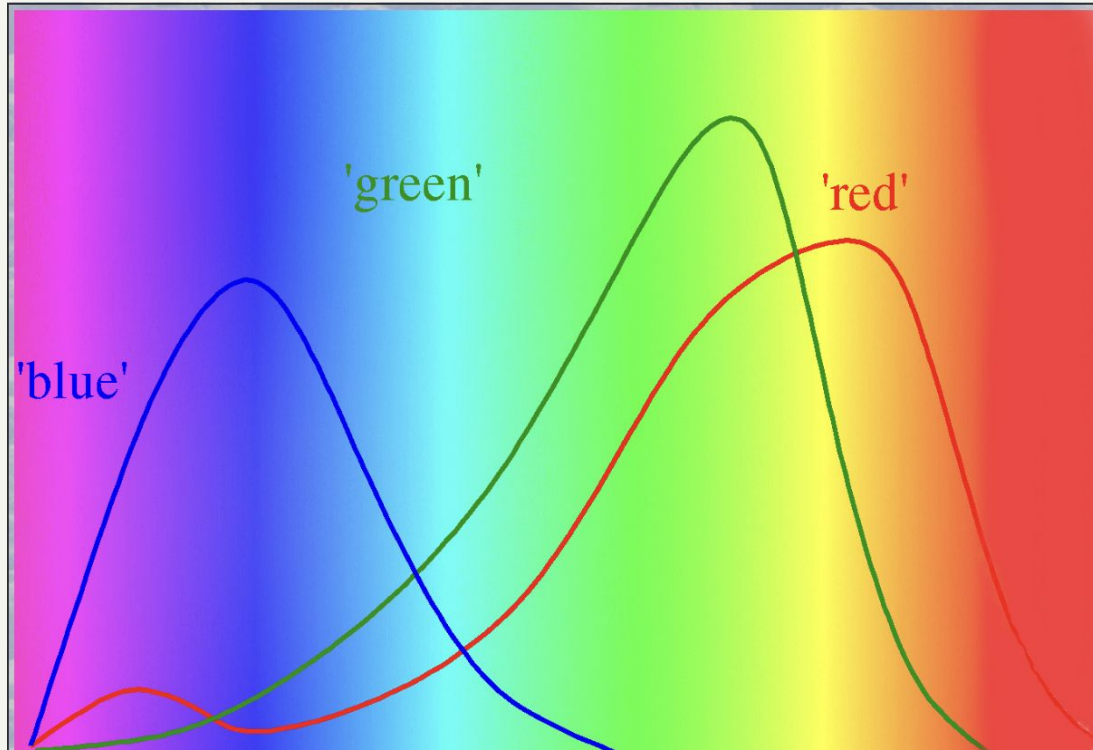
Cones and Rods in the Human Retina

The human retina contains approximately **6 million cones** and around **100 million rods**. Cones are responsible for color vision and are divided into three types based on their sensitivity to different wavelengths of light:

- **L-cones** (Long-wavelength sensitive) – respond to **red light** (~65% of all cones).
- **M-cones** (Medium-wavelength sensitive) – respond to **green light** (~33%).
- **S-cones** (Short-wavelength sensitive) – respond to **blue light** (~2%).

Humans, like some other primates, have **trichromatic vision**, meaning they can perceive colors through the combination of signals from these three types of cones.

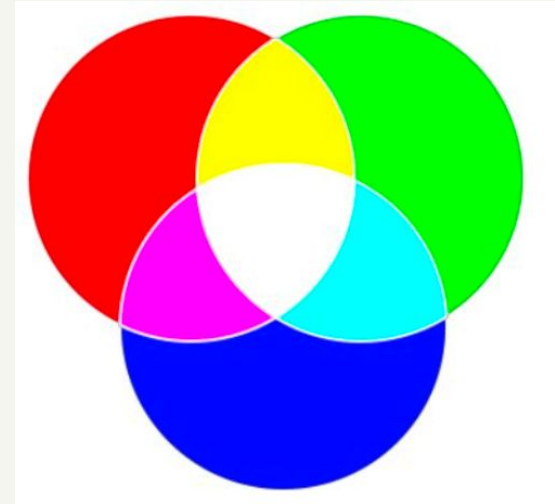
Color Perception



Primary Colors

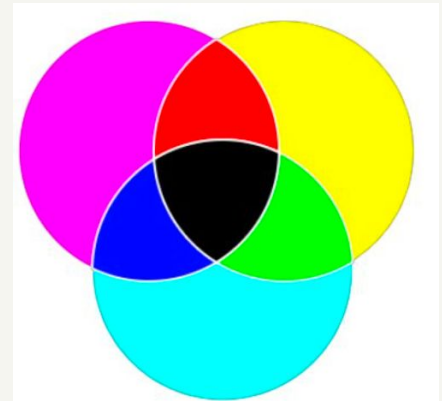
Primary Colors of Light: RGB

- **Red, Green, Blue (RGB)** are the three **primary colors of light** used in digital screens and other light-emitting technologies.
- This system follows **additive color mixing**, meaning colors are created by **adding light** of different wavelengths.
- When **all three colors are combined at full intensity**, they produce **white light**.



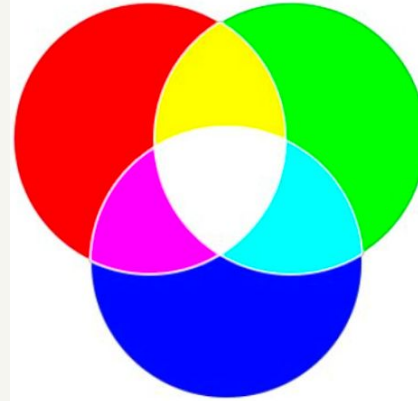
Primary Colors of Pigments: CMYK

- **Cyan, Magenta, Yellow, and Black (CMYK)** are the **primary colors of pigments and inks** used in printing.
- This system follows **subtractive color mixing**, meaning colors are created by **absorbing (subtracting) certain wavelengths of light** and reflecting others.
- When **all three main colors (CMY)** are **mixed together**, they ideally create **black**, but in practice, they produce a dark brownish color.

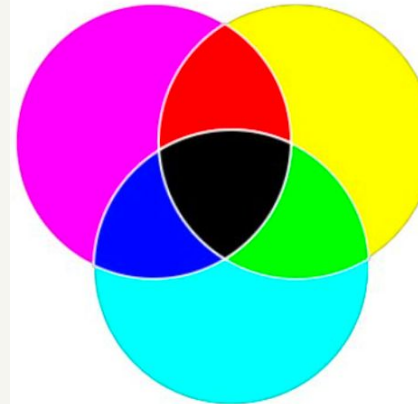


Key Differences

- **RGB (Additive Mixing):** Adding colors together increases brightness, used in devices that emit light.
- **CMYK (Subtractive Mixing):** Adding pigments together darkens the result, used in materials that reflect light.



Mixtures of light



Mixtures of pigments

Colors on Displays

How Displays Create Images

Modern digital screens—such as those on TVs, computer monitors, and smartphones—create images using **light emission**. This process is based on the **RGB (Red, Green, Blue) color model**, which follows **additive color mixing**.

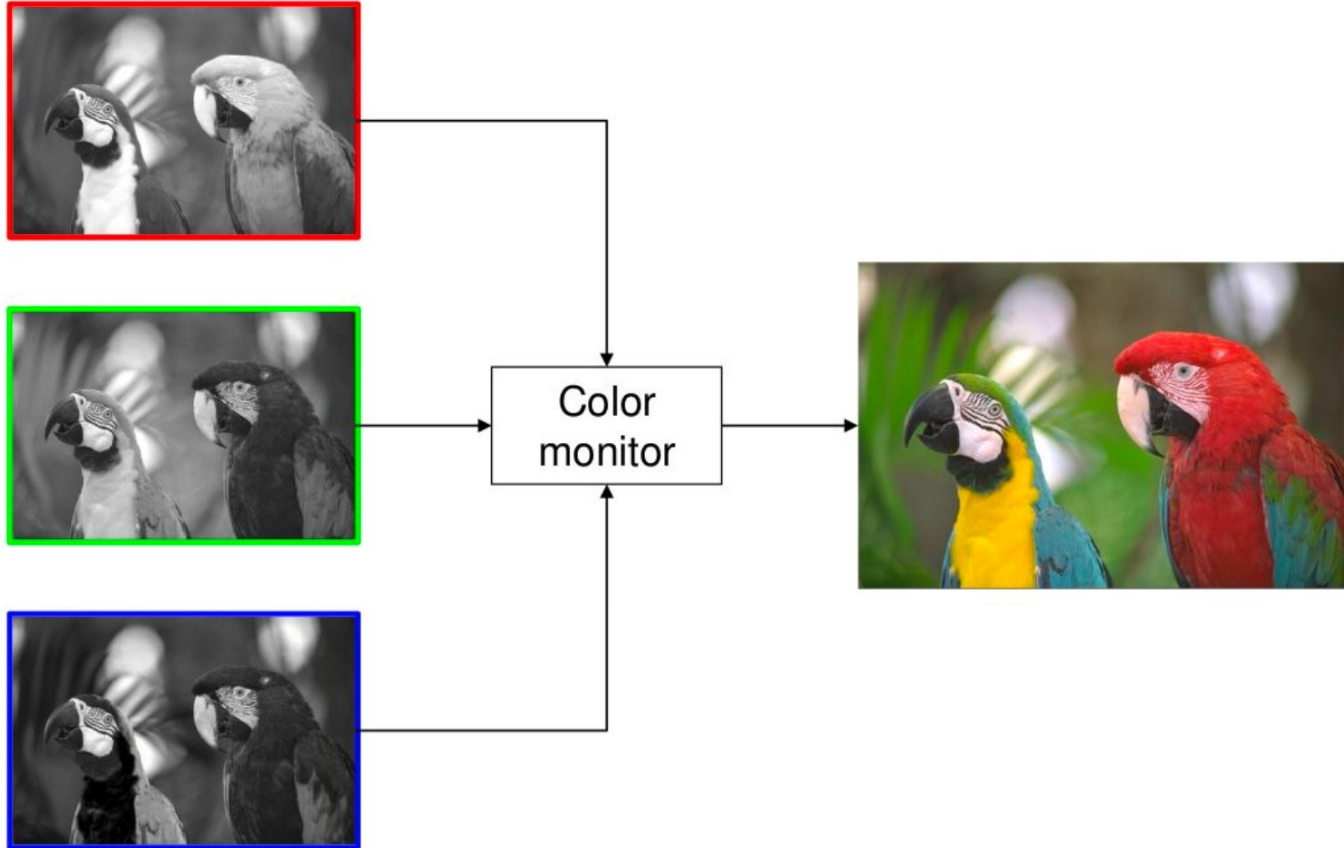


Additive Color Mixing

- **Full intensity of all three colors (R + G + B) → White**
- **No light emitted (0 intensity for all) → Black**
- **Equal intensity of two colors → A new secondary color:**
 - **Red + Green → Yellow**
 - **Green + Blue → Cyan**
 - **Red + Blue → Magenta**

By adjusting the brightness of each subpixel, screens can create **millions of colors**.

Additive Color Mixing



Why RGB is used instead of CMYK?

- Screens **emit light**, so additive RGB mixing works best.
- CMYK is designed for **printed materials**, where colors are formed by subtracting light from white paper.

Storing Images

Pixel Matrices and Color Channels

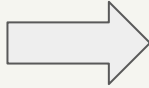
Digital images are stored as **matrices of pixels**, where each pixel represents a tiny portion of the image. The way these pixels are stored depends on whether the image is **grayscale** (black and white) or **colorful**.



Grayscale Images

- Grayscale images contain **only one channel**, where each pixel is represented by a **single intensity value**.
- The value typically ranges from **0 (black)** to **255 (white)** in an **8-bit image**.
- The image is stored as a **2D matrix**, where each cell represents a pixel's brightness.

Grayscale Images



```
[[ 0, 50, 100, 150, 200],  
 [25, 75, 125, 175, 225],  
 [50, 100, 150, 200, 250],  
 [75, 125, 175, 225, 255],  
 [100, 150, 200, 250, 255]]
```

Darker shades have **lower values**, while brighter shades have **higher values**.

Color Images

- Color images typically use the **RGB (Red, Green, Blue) model**, meaning each pixel has **three color values**.
- Each pixel is stored as a **set of three numbers**, one for each color channel.
- The image is stored as a **3D matrix** (Height $\times$ Width $\times$ 3), where each layer represents one color channel.

Color Images

Examples of a color pixels in RGB format

`[255, 0, 0]` → Pure Red

`[0, 255, 0]` → Pure Green

`[0, 0, 255]` → Pure Blue

`[255, 255, 255]` → White

Each line represents only **one** pixel!

Color Images



```
[[ 29, 35, 41, 40, 37, 38, 37, 43, 38, 42],  
[ 43, 36, 41, 37, 43, 37, 45, 48, 47, 43],  
[[124, 132, 145, 143, 140, 137, 134, 140, 138, 138], 32],  
[132, 129, 136, 135, 140, 135, 140, 143, 145, 141], 42],  
[[ 95, 103, 112, 113, 110, 110, 108, 113, 109, 109], 129], 43],  
[104, 100, 107, 106, 113, 106, 111, 114, 115, 111], 136], 37],  
[100, 101, 108, 104, 113, 107, 109, 114, 100, 102], 135], 42],  
[ 97, 98, 97, 100, 104, 104, 103, 103, 94, 110], 127], 45],  
[ 86, 90, 90, 90, 91, 94, 92, 98, 99, 109], 126], 33],  
[ 74, 89, 90, 88, 99, 96, 97, 90, 96, 99], 128], 31],  
[ 72, 80, 78, 81, 86, 90, 93, 95, 96, 99], 110],  
[ 65, 69, 75, 81, 80, 85, 82, 90, 96, 99], 98],  
[133, 102, 91, 85, 74, 80, 80, 80, 78, 82],  
[203, 185, 173, 134, 82, 62, 66, 75, 76, 71]]
```

- Each pixel contains **three values** (Red, Green, Blue).
- Mixing different intensities creates different colors.

<Coding time>

Q&A time!